Engineering Performance Task — Tabletop Alert / Signal Device

Chapter 6 · Energy Transformations + Engineering Design · Week 4

Name: _____ Date: _____

THE PROBLEM

Design a non-electrical tabletop device that produces a clear signal when a model door/gate/trigger moves. Your device must demonstrate at least one energy transformation.

CRITERIA (what a good device must do)

- ☐ 1. responds to the trigger (the model door/gate moves → the device reacts)
- ☐ 2. produces an observable signal (sound, motion, or a visible signal)
- ☐ 3. demonstrates at least one energy transformation
- ☐ 4. works repeatedly (at least 2 of 3 trials)
- ☐ 5. you can explain the energy pathway

CONSTRAINTS (limits)

- ☐ 1. only the provided materials
- ☐ 2. no wall electricity · no open flame · no hot plate · no projectile launching
- ☐ 3. limited build time (one period per build/test)
- ☐ 4. limited material quantity
- ☐ 5. tabletop footprint (fits on the tray / sheet of paper)

MY TEAM'S PROBLEM (in our own words)

DEVICE NAME

MATERIALS I WILL USE

Circle or list: cardboard tubes · paper cups · craft sticks · rubber bands · string · paper clips · washers · bells / metal cup · index cards / paper · tape · glue · blocks · foil

BLUEPRINT (draw + label)

Label: the TRIGGER, the OUTPUT, and where the ENERGY CHANGE happens.

[draw + label your blueprint here]

HOW I WILL KNOW IT WORKS

MY ENERGY PATHWAY

INPUT → CHANGE IN FORM (TRANSFORMATION) → OUTPUT / OBSERVABLE CHANGE

INPUT (energy at the start): _____

TRANSFORMATION (what changes form): _____

OUTPUT (what I can observe): _____

TRANSFER / TRANSFORMATION / BOTH?

Does my device transfer energy, transform energy, or BOTH? Circle one and explain.

TRANSFER TRANSFORMATION BOTH

TEST 1 — record your evidence (3 trials)

Trial	Did it trigger? (yes/no)	What output occurred?	Evidence (what I observed)
1			
2			
3			

WHAT WORKED / WHAT FAILED

What worked: _____

What failed (this is DATA, not a penalty): _____

Evidence I collected: _____

REDESIGN — change ONE thing, based on evidence

WHAT WE CHANGED (only one thing): _____

WHY (what evidence from Test 1 supports this change): _____

PREDICTION (I think Test 2 will show): _____

NEW BLUEPRINT SKETCH

[sketch the change]

TEST 2 — same conditions as Test 1 (3 trials)

Trial	Did it trigger? (yes/no)	What output occurred?	Evidence (what I observed)
1			
2			
3			

TEST 1 vs TEST 2 — compare

Test 1 result: _____

Test 2 result: _____

Did the change help? (yes / no / same)

Evidence that supports my answer: _____

FINAL EXPLANATION (claim · evidence · reasoning)

CLAIM (what my device does): _____

EVIDENCE (what I observed): _____

REASONING (why the evidence supports my claim): _____

MY ENERGY PATHWAY (final)

INPUT → TRANSFORMATION → OUTPUT:

_____ → _____ → _____

PRESENTATION PLAN

Be ready to say: the problem · the criteria · the constraints · the blueprint · Test 1 · the redesign · Test 2 · the energy pathway · the final evaluation.

My one-sentence summary: _____

WHAT I LEARNED